SOCIOL 723: Social Statistics II

Week 14, Synthesis and Exam Review

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Today

- A synthesis: what the semester was actually about.
- A decision guide you can use on your own data.
- The final exam: what it covers, how it is run, and how to prepare.

Logistics

No lab Thursday November 26 (Thanksgiving recess).
Cumulative final exam: date and time TBD, set by class poll.

Synthesis

Three Questions, in Order ★

1. **What am I trying to estimate?** A unit-specific quantity and a target population. ATE, ATT, LATE, the effect at a cutoff, $\tau(x)$. Weeks 5 and 13.
2. **What would have to be true for my estimate to recover it?** Ignorability, exclusion, continuity, parallel trends. Weeks 5–11.
3. **How do I estimate it, and how uncertain am I?** Least squares, maximum likelihood, weighting, orthogonal scores, cross-fitting. Weeks 1–4, 6, 12–13.

The ordering is the lesson

Most quantitative papers start at 3 and never articulate 1 or 2. Most methodological disputes are really disagreements at 1 or 2 being conducted as if they were about 3.

One Idea, Recurring

FWL (Wk 1)
residualize D on W

Weighting / AIPW (Wk
residualize, then reweight

Flexible nuisances (Wk 13)
the same idea, fewer
functional-form claims

- **Partialling out** is the through-line. Remove the part of the treatment explained by the controls; use what is left.
- Its virtue is *insensitivity*: small errors in how you model the controls do not contaminate the parameter you care about.
- Everything else, matching, weighting, IV, RD, DID, synthetic control , is a different answer to the question *where does the counterfactual come from?*

Every Design, in One Table ★

Design	Key assumption	Estiman
Regression / matching / IPW	ignorability given X	ATE, A
AIPW / DML	ignorability + rates	ATE, $\tau($
IV / 2SLS	relevance, independence, exclusion, monotonicity	LATE (
RD (sharp)	continuity at c	effect at
RD (fuzzy)	+ first-stage jump, monotonicity, exclusion	complie
DID (2×2)	parallel trends, no anticipation	ATT
DID, staggered	cohort-wise parallel trends vs. not-yet-treated, no anticipation	ATT$(g,$
Synthetic control	factor model, long pre-period	ATT fo

Every Design, in One Table ★ (cont.)

- No row is safer than the others in the abstract. Credibility is a property of the application, not of the method.
- Notice how often the estimand is chosen *by the design* rather than by the researcher. Say so when it is.

Inference, in One Table ★

Situation	Use	Week
Default	HC2 or HC3 robust	2
Treatment assigned in groups	cluster on the assignment level	2
Fewer than ∼40 clusters	CR2 + Satterthwaite, or wild cluster bootstrap	2
Nonlinear quantity of interest	delta method, simulation, or bootstrap	2, 4
Nuisance function estimated	bootstrap the whole pipeline, or an influence function	6
Weak instrument	Anderson–Rubin	7
One treated unit	permutation / placebo inference	11
ML nuisances	orthogonal score + cross-fitting	13

The recurring error is a standard error that ignores a step in the procedure, estimated weights, a selected model, a matched design, a chosen bandwidth. When in doubt,

Inference, in One Table ★ (cont.)

bootstrap the whole pipeline, resampling in a way that *reproduces the sampling or assignment structure* (clusters as clusters, panels as panels). But the bootstrap is not universal: it is *invalid* for nearest-neighbour matching (Week 6), and unreliable more generally for non-smooth, boundary, post-selection, and weakly identified procedures.

A Decision Guide ★

1. Write the theoretical estimand in one sentence: unit-specific quantity, target population.
2. Draw the DAG. Mark what is unobserved.
3. Ask: *is there any set I can measure that closes the back doors?*
   - Yes → selection on observables. Check overlap, use a doubly robust estimator, report a sensitivity analysis (Week 6).
   - No → look for design-based variation: an instrument (7), a threshold (9), a policy change over time (10), a single treated unit with a long pre-period (11).
4. Check whether the design's estimand is the one you wrote in step 1. If not, say so explicitly rather than quietly redefining the question.
5. Choose inference to match the design.
6. If W is high-dimensional or the functional form is a guess, use an orthogonal score with cross-fitting (13).
7. Report what would change your mind.

Things That Are Never a Solution ★

- Adding more controls without a DAG. Bad controls exist, and "kitchen sink" amplifies bias as often as it reduces it (Week 5).
- Robust standard errors for a problem of bias (Week 2).
- A high R^2, or a high AUC, as evidence of causality (Weeks 1, 12).
- A non-significant specification test, balance t-tests, Sargan, pre-trend tests, as evidence that an assumption holds. Low power is not confirmation (Weeks 6, 7, 10).
- Machine learning for an identification problem (Week 13).
- Reporting whichever specification worked. Pre-commit, then show robustness.

Each of these is a real move made in published sociology, and you now have the vocabulary to say precisely why it fails.

Final Exam

What the Exam Covers

Cumulative, but weighted towards Weeks 9–13: the midterm already examined Weeks 1–7, and this is the only assessment that reaches the second half of the course.

- **Everything developed in scalar form is examinable.** Frames marked ★ are the essential spine, study those first. Frames marked ★ are not examinable: matrix derivations, projection geometry, asymptotic arguments.
- For every design, you should be able to state *what it assumes*, *what it estimates*, and *how it fails*. That is the table from earlier today, and it is the spine of the exam.
- Expect to read a short description of a study and say which design it is, what its estimand is, and what you would need to believe.
- No code. You may be asked to interpret `R` output.

Format and Rules

Logistics

Date and time **to be determined**: we will poll the class and announce well in advance. Open book and open note, but **timed**. The exam counts 35%, or 25% if you elected the **project option**, in which case your 6–10 page project memo and replication code are due Friday, December 11.

No AI

Unlike the problem sets, no AI assistance of any kind. This is deliberate. It is the one place in the course where what is being assessed is unambiguously *your* understanding rather than what a tool can produce for you.

Bring the slides and your own notes. Notes you wrote yourself are far more useful under time pressure than a printout you have never read.

How to Prepare

1. **Work the blue stars.** Go deck by deck through the frames marked ★ and check you can explain and use each one without notes. If you can, you are ready.
2. **Rebuild the four tables** from today: design, assumption, estimand, threat; and the inference table. Write them from memory, then check.
3. **Re-derive the short results by hand**: the normal equations, the scalar variance, omitted-variable bias, the Wald ratio, the 2×2 DID. None takes more than a few lines.
4. **Re-read your own problem sets**, especially anything you got wrong. That is the highest-yield hour you can spend.

The exam rewards judgement, not recall. If you can say why a design is the wrong one for a question, you understand it.

Where to Go Next

- **Depth on design**: Angrist and Pischke, *Mostly Harmless Econometrics*; Cunningham, *Causal Inference: The Mixtape*; Huntington-Klein, *The Effect*.
- **Depth on potential outcomes**: Imbens and Rubin, *Causal Inference for Statistics, Social, and Biomedical Sciences*; Morgan and Winship.
- **Depth on graphs**: Pearl, *Causality*; Hernán and Robins, *Causal Inference: What If*.
- **Depth on causal ML**: Chernozhukov et al., *Applied Causal Inference Powered by ML and AI*.
- **In this department**: measurement and latent variables, computational text analysis, network methods, demographic methods, multilevel models, each of which assumes what you now know.

Where to Go Next (cont.)

The one thing to keep

Say what you are estimating, say what would have to be true, and say what would change your mind. Everything else is technique.